

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 100.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 384

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 32

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 209866141261.20148

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 249997.98038142629

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 1349869.3732390273

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.00214463576

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.4999992875539916

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 1.364865356190631

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 475532.63962963567

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 0.0857

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 1.0000

Parameter bounds [1.000E+000; 1.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 2.0000

Parameter bounds [2.000E+000; 2.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.9989

Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999992756.0958087

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 115573.05694589266

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 9578216864939.172

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 221219.44656305094

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 0.9165096460629828

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 5.100546695066028E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.1081671389281706E-12

Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999984874.2064532

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 89330.02443982993

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 5047692073445.518

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 159291.60020812045

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 1.693493142057753

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 1.0205123651060284E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.3371173995054896E-12

Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 999985605.8911237

Parameter bounds [1.000E+000; 1.000E+009]

Activation energy for condensed phase reactions 112352.17466605104

Parameter bounds [5.000E+004; 3.000E+005]

Pre-exponential factor for gas-phase reactions within skeleton structure 33803903.26161451

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions within skeleton structure 50000.0058806014

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in skeleton structure 0.9779554138177717

Parameter bounds [0.000E+000; 2.500E+000]

Coefficient A for metal combustion heat flux density in skeleton structure, 5.361720037036252E-07

Parameter bounds [1.000E-010; 1.000E-003]

Coefficient B for metal combustion heat flux density in skeleton structure, 1.0126937397623123E-12

Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.2073

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 9.49 mm/s

First Conditional Fuel

Surface temperature 757.25 K

Linear burning rate 10.72 mm/s

Mass decomposition rate 17.79 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.07e+07 W/m²

Kinetic flame height 14.24 μm
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature 2,232.13 K
Binder sublimation heat flux density 2.07e+07 W/m^2
Surface heat balance error -0.0041 W/m^2

Secondary Conditional Fuel

Surface temperature 722.84 K
Linear burning rate 4.59 mm/s
Mass decomposition rate 7.61 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.15e+06 W/m^2
Kinetic flame height 76.36 μm
Average kinetic flame density 2.57 kg/m^3
Average kinetic flame temperature 1,543.92 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 7.12e+06 W/m^2
Kinetic flame height 12.19 μm
Average kinetic flame density 3.43 kg/m^3
Average kinetic flame temperature 1,156.92 K
Heat flux density due to metal combustion, 1.99e+06 W/m^2
Effective thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 0.4 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 7.3e-06 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.17e-04 m
Pore diameter 4.82e-08 m

Diffusion flame heat flux density 3.82e+06 W/m^2
Diffusion flame height 75.51 μm
Heat flux density to surface within skeletal structure 3.27e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.38e+06 W/m^2
Total heat flux density to surface 8.47e+06 W/m^2
Binder sublimation heat flux density 8.47e+06 W/m^2
Surface heat balance error -0.0012 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 18 mm/s

First Conditional Fuel

Surface temperature 841.44 K
Linear burning rate 63.94 mm/s
Mass decomposition rate 106.07 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 1.37e+08 W/m^2
Kinetic flame height 2.09 μm
Average kinetic flame density 7.08 kg/m^3
Average kinetic flame temperature 2,274.22 K
Binder sublimation heat flux density 1.37e+08 W/m^2
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 743.62 K
Linear burning rate 7.73 mm/s
Mass decomposition rate 12.83 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 8.66e+06 W/m^2
Kinetic flame height 18.73 μm
Average kinetic flame density 10.36 kg/m^3
Average kinetic flame temperature 1,554.31 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.68e+07 W/m^2
Kinetic flame height 5.04 μm

Average kinetic flame density 13.79 kg/m^3
Average kinetic flame temperature $1,167.31 \text{ K}$
Heat flux density due to metal combustion, $3.24\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $2.79\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $6.94\text{e}-05 \text{ m}$
Pore diameter $1.69\text{e}-08 \text{ m}$
Diffusion flame heat flux density $2.25\text{e}+06 \text{ W/m}^2$
Diffusion flame height $127.31 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $6.63\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $5.8\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.47\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.47\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0015 W/m^2

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 22.71 mm/s

First Conditional Fuel

Surface temperature 874.3 K
Linear burning rate 116.92 mm/s
Mass decomposition rate $193.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.6\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.09 \text{ }\mu\text{m}$
Average kinetic flame density 11.26 kg/m^3
Average kinetic flame temperature $2,290.65 \text{ K}$
Binder sublimation heat flux density $2.6\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.033 W/m^2

Secondary Conditional Fuel

Surface temperature 752.58 K
Linear burning rate 9.6 mm/s
Mass decomposition rate $15.93 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.33\text{e}+07 \text{ W/m}^2$
Kinetic flame height $12.13 \text{ }\mu\text{m}$
Average kinetic flame density 16.55 kg/m^3
Average kinetic flame temperature $1,558.79 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.16\text{e}+07 \text{ W/m}^2$
Kinetic flame height $3.88 \text{ }\mu\text{m}$
Average kinetic flame density 22.02 kg/m^3
Average kinetic flame temperature $1,171.79 \text{ K}$
Heat flux density due to metal combustion, $3.96\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.88\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $5.59\text{e}-05 \text{ m}$
Pore diameter $1.1\text{e}-08 \text{ m}$
Diffusion flame heat flux density $1.8\text{e}+06 \text{ W/m}^2$
Diffusion flame height $158.08 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $6.95\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $9.68\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.84\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.84\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0011 W/m^2

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 14.32 mm/s

First Conditional Fuel

Surface temperature 757.25 K
Linear burning rate 10.72 mm/s
Mass decomposition rate $17.79 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.07\text{e}+07 \text{ W/m}^2$
Kinetic flame height 14.24 μm
Average kinetic flame density 1.78 kg/m^3
Average kinetic flame temperature 2,232.13 K
Binder sublimation heat flux density $2.07\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0041 W/m^2

Secondary Conditional Fuel

Surface temperature 757.25 K
Linear burning rate 10.72 mm/s
Mass decomposition rate $17.79 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $925,726.33 \text{ W/m}^2$
Kinetic flame height 173.67 μm
Average kinetic flame density 2.54 kg/m^3
Average kinetic flame temperature 1,561.13 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.33\text{e}+07 \text{ W/m}^2$
Kinetic flame height 8.01 μm
Average kinetic flame density 2.35 kg/m^3
Average kinetic flame temperature 1,690.13 K
Heat flux density due to metal combustion, $2.41\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $2.12\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $5\text{e}-05 \text{ m}$
Pore diameter $8.81\text{e}-09 \text{ m}$
Diffusion flame heat flux density $1.2\text{e}+07 \text{ W/m}^2$
Diffusion flame height 24.22 μm
Heat flux density to surface within skeletal structure $8.04\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $636,056.86 \text{ W/m}^2$
Total heat flux density to surface $2.07\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.07\text{e}+07 \text{ W/m}^2$
Surface heat balance error $-5.54\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 29.22 mm/s

First Conditional Fuel

Surface temperature 841.44 K
Linear burning rate 63.94 mm/s
Mass decomposition rate $106.07 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.37\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.09 μm
Average kinetic flame density 7.08 kg/m^3
Average kinetic flame temperature 2,274.22 K
Binder sublimation heat flux density $1.37\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 770.67 K

Linear burning rate 14.63 mm/s

Mass decomposition rate $24.27 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $4.6\text{e}+06 \text{ W/m}^2$

Kinetic flame height 34.68 μm

Average kinetic flame density 10.27 kg/m^3

Average kinetic flame temperature 1,567.83 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.73\text{e}+07 \text{ W/m}^2$

Kinetic flame height 2.75 μm

Average kinetic flame density 9.49 kg/m^3

Average kinetic flame temperature 1,696.83 K

Heat flux density due to metal combustion, $3.2\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $1.2\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness $3.66\text{e}-05 \text{ m}$

Pore diameter $4.73\text{e}-09 \text{ m}$

Diffusion flame heat flux density $8.79\text{e}+06 \text{ W/m}^2$

Diffusion flame height 33.05 μm

Heat flux density to surface within skeletal structure $1.64\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $3.52\text{e}+06 \text{ W/m}^2$

Total heat flux density to surface $2.87\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $2.87\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0031 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s

Calculated burning rate 33.25 mm/s

First Conditional Fuel

Surface temperature 874.3 K

Linear burning rate 116.92 mm/s

Mass decomposition rate $193.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.6\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.09 μm

Average kinetic flame density 11.26 kg/m^3

Average kinetic flame temperature 2,290.65 K

Binder sublimation heat flux density $2.6\text{e}+08 \text{ W/m}^2$

Surface heat balance error -0.033 W/m^2

Secondary Conditional Fuel

Surface temperature 773.57 K

Linear burning rate 15.63 mm/s

Mass decomposition rate $25.92 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $8.2\text{e}+06 \text{ W/m}^2$

Kinetic flame height 19.41 μm

Average kinetic flame density 16.44 kg/m^3

Average kinetic flame temperature 1,569.29 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.85 μm

Average kinetic flame density 15.19 kg/m^3

Average kinetic flame temperature 1,698.29 K

Heat flux density due to metal combustion, $3.4\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $1.06\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness $3.43\text{e-}05$ m
Pore diameter $4.15\text{e-}09$ m
Diffusion flame heat flux density $8.22\text{e+}06$ W/m²
Diffusion flame height 35.3 μm
Heat flux density to surface within skeletal structure $1.56\text{e+}07$ W/m²
Heat flux density to surface outside skeletal structure $6.96\text{e+}06$ W/m²
Total heat flux density to surface $3.08\text{e+}07$ W/m²
Binder sublimation heat flux density $3.08\text{e+}07$ W/m²
Surface heat balance error 0.0059 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 10.74 mm/s

First Conditional Fuel

Surface temperature 757.25 K
Linear burning rate 10.72 mm/s
Mass decomposition rate 17.79 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface $2.07\text{e+}07$ W/m²
Kinetic flame height 14.24 μm
Average kinetic flame density 1.78 kg/m³
Average kinetic flame temperature $2,232.13$ K
Binder sublimation heat flux density $2.07\text{e+}07$ W/m²
Surface heat balance error -0.0041 W/m²

Secondary Conditional Fuel

Surface temperature 715.64 K
Linear burning rate 3.8 mm/s
Mass decomposition rate 6.3 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $58,393.66$ W/m²
Kinetic flame height 558.9 μm
Average kinetic flame density 4.52 kg/m³
Average kinetic flame temperature 878.82 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.37\text{e+}06$ W/m²
Kinetic flame height 22.15 μm
Average kinetic flame density 4.06 kg/m³
Average kinetic flame temperature 978.32 K
Heat flux density due to metal combustion, $5.84\text{e+}06$ W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity $8.23\text{e-}06$ W/(m·K)
Conductive thermal conductivity balance error $9.240271880983641\text{E-}10$
Skeleton layer thickness $1.41\text{e-}04$ m
Pore diameter $7.02\text{e-}08$ m
Diffusion flame heat flux density $2.58\text{e+}06$ W/m²
Diffusion flame height 111.21 μm
Heat flux density to surface within skeletal structure $4.34\text{e+}06$ W/m²
Heat flux density to surface outside skeletal structure $27,516.98$ W/m²
Total heat flux density to surface $6.95\text{e+}06$ W/m²
Binder sublimation heat flux density $6.95\text{e+}06$ W/m²
Surface heat balance error $-5.56\text{e-}04$ W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 17.75 mm/s

First Conditional Fuel

Surface temperature 841.44 K

Linear burning rate 63.94 mm/s
Mass decomposition rate $106.07 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.37\text{e}+08 \text{ W/m}^2$
Kinetic flame height 2.09 μm
Average kinetic flame density 7.08 kg/m^3
Average kinetic flame temperature 2,274.22 K
Binder sublimation heat flux density $1.37\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 727.95 K
Linear burning rate 5.23 mm/s
Mass decomposition rate $8.67 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $286,682.17 \text{ W/m}^2$
Kinetic flame height 109.55 μm
Average kinetic flame density 18.19 kg/m^3
Average kinetic flame temperature 884.97 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.84\text{e}+06 \text{ W/m}^2$
Kinetic flame height 7.5 μm
Average kinetic flame density 16.35 kg/m^3
Average kinetic flame temperature 984.47 K
Heat flux density due to metal combustion, $7.87\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $4.57\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $1.03\text{e}-04 \text{ m}$
Pore diameter $3.7\text{e}-08 \text{ m}$

Diffusion flame heat flux density $1.86\text{e}+06 \text{ W/m}^2$
Diffusion flame height 153.06 μm
Heat flux density to surface within skeletal structure $7.72\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $136,179.86 \text{ W/m}^2$
Total heat flux density to surface $9.72\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $9.72\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-9.96\text{e}-06 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 21.17 mm/s

First Conditional Fuel

Surface temperature 874.3 K
Linear burning rate 116.92 mm/s
Mass decomposition rate $193.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.6\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.09 μm
Average kinetic flame density 11.26 kg/m^3
Average kinetic flame temperature 2,290.65 K
Binder sublimation heat flux density $2.6\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.033 W/m^2

Secondary Conditional Fuel

Surface temperature 734.02 K
Linear burning rate 6.1 mm/s
Mass decomposition rate $10.11 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $467,595.95 \text{ W/m}^2$
Kinetic flame height 65.86 μm
Average kinetic flame density 29.05 kg/m^3
Average kinetic flame temperature 888.01 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $9.34\text{e}+06 \text{ W/m}^2$
Kinetic flame height $5.43 \mu\text{m}$
Average kinetic flame density 26.12 kg/m^3
Average kinetic flame temperature 987.51 K
Heat flux density due to metal combustion, $9.08\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $3.45\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $8.79\text{e}-05 \text{ m}$
Pore diameter $2.72\text{e}-08 \text{ m}$
Diffusion flame heat flux density $1.59\text{e}+06 \text{ W/m}^2$
Diffusion flame height $178.47 \mu\text{m}$
Heat flux density to surface within skeletal structure $9.61\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $223,622.07 \text{ W/m}^2$
Total heat flux density to surface $1.14\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.14\text{e}+07 \text{ W/m}^2$
Surface heat balance error $4.31\text{e}-04 \text{ W/m}^2$

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 17.28 mm/s

First Conditional Fuel

Surface temperature 600 K
Linear burning rate 10.08 mm/s
Mass decomposition rate $16.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.55\text{e}+07 \text{ W/m}^2$
Kinetic flame height $20.01 \mu\text{m}$
Average kinetic flame density 1.84 kg/m^3
Average kinetic flame temperature $2,153.5 \text{ K}$
Binder sublimation heat flux density $1.55\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0026 W/m^2

Secondary Conditional Fuel

Surface temperature 600 K
Linear burning rate 10.08 mm/s
Mass decomposition rate $16.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $945,843.51 \text{ W/m}^2$
Kinetic flame height $186.61 \mu\text{m}$
Average kinetic flame density 2.68 kg/m^3
Average kinetic flame temperature $1,482.5 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.72\text{e}+06 \text{ W/m}^2$
Kinetic flame height $14.74 \mu\text{m}$
Average kinetic flame density 3.62 kg/m^3
Average kinetic flame temperature $1,095.5 \text{ K}$
Heat flux density due to metal combustion, $2.79\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.14\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.01\text{e}-05 \text{ m}$
Pore diameter $1.32\text{e}-08 \text{ m}$
Diffusion flame heat flux density $1.85\text{e}+06 \text{ W/m}^2$
Diffusion flame height $165.97 \mu\text{m}$
Heat flux density to surface within skeletal structure $1.31\text{e}+07 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 588,263.31 W/m²
Total heat flux density to surface 1.55e+07 W/m²
Binder sublimation heat flux density 1.55e+07 W/m²
Surface heat balance error 8.37e-04 W/m²

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 32.92 mm/s

First Conditional Fuel

Surface temperature 666.02 K
Linear burning rate 59.48 mm/s
Mass decomposition rate 98.67 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.01e+08 W/m²
Kinetic flame height 3 µm
Average kinetic flame density 7.36 kg/m³
Average kinetic flame temperature 2,186.51 K
Binder sublimation heat flux density 1.01e+08 W/m²
Surface heat balance error 0.012 W/m²

Secondary Conditional Fuel

Surface temperature 613.35 K
Linear burning rate 14.88 mm/s
Mass decomposition rate 24.69 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.35e+06 W/m²
Kinetic flame height 40.28 µm
Average kinetic flame density 10.81 kg/m³
Average kinetic flame temperature 1,489.17 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.29e+07 W/m²
Kinetic flame height 1.85 µm
Average kinetic flame density 14.6 kg/m³
Average kinetic flame temperature 1,102.17 K
Heat flux density due to metal combustion, 4.04e+07 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 5.6e-07 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.86e-06 m
Pore diameter 6.04e-09 m
Diffusion flame heat flux density 1.25e+06 W/m²
Diffusion flame height 245.08 µm
Heat flux density to surface within skeletal structure 1.87e+07 W/m²
Heat flux density to surface outside skeletal structure 3.48e+06 W/m²
Total heat flux density to surface 2.34e+07 W/m²
Binder sublimation heat flux density 2.34e+07 W/m²
Surface heat balance error 0.0026 W/m²

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s
Calculated burning rate 43.54 mm/s

First Conditional Fuel

Surface temperature 691.75 K
Linear burning rate 108.36 mm/s
Mass decomposition rate 179.76 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.92e+08 W/m²
Kinetic flame height 1.57 µm
Average kinetic flame density 11.73 kg/m³
Average kinetic flame temperature 2,199.37 K
Binder sublimation heat flux density 1.92e+08 W/m²

Surface heat balance error -0.022 W/m²

Secondary Conditional Fuel

Surface temperature 622.26 K

Linear burning rate 19.13 mm/s

Mass decomposition rate 31.73 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 6.46e+06 W/m²

Kinetic flame height 26.97 μm

Average kinetic flame density 17.27 kg/m³

Average kinetic flame temperature 1,493.63 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 9.66e+07 W/m²

Kinetic flame height 1 μm

Average kinetic flame density 23.31 kg/m³

Average kinetic flame temperature 1,106.63 K

Heat flux density due to metal combustion, 5.13e+07 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 3.54e-07 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 5.34e-06 m

Pore diameter 3.66e-09 m

Diffusion flame heat flux density 966,451.27 W/m²

Diffusion flame height 314.94 μm

Heat flux density to surface within skeletal structure 2.41e+07 W/m²

Heat flux density to surface outside skeletal structure 5.41e+06 W/m²

Total heat flux density to surface 3.05e+07 W/m²

Binder sublimation heat flux density 3.05e+07 W/m²

Surface heat balance error 0.0016 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 8.05 mm/s

First Conditional Fuel

Surface temperature 779.36 K

Linear burning rate 10.82 mm/s

Mass decomposition rate 17.95 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.15e+07 W/m²

Kinetic flame height 13.62 μm

Average kinetic flame density 1.77 kg/m³

Average kinetic flame temperature 2,243.18 K

Binder sublimation heat flux density 2.15e+07 W/m²

Surface heat balance error -0.0026 W/m²

Secondary Conditional Fuel

Surface temperature 735.94 K

Linear burning rate 3.78 mm/s

Mass decomposition rate 6.27 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.62e+06 W/m²

Kinetic flame height 62.25 μm

Average kinetic flame density 2.56 kg/m³

Average kinetic flame temperature 1,550.47 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 47,097.32 W/m²

Kinetic flame height 1,815.52 μm

Average kinetic flame density 3.41 kg/m³

Average kinetic flame temperature 1,163.47 K

Heat flux density due to metal combustion, $1.69\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.24\text{e}-05 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
Pore diameter $7.76\text{e}-08 \text{ m}$
Diffusion flame heat flux density $4.71\text{e}+06 \text{ W/m}^2$
Diffusion flame height $62.22 \mu\text{m}$
Heat flux density to surface within skeletal structure $448,259.57 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.94\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $7.1\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $7.1\text{e}+06 \text{ W/m}^2$
Surface heat balance error $6.51\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s
Calculated burning rate 14.71 mm/s

First Conditional Fuel

Surface temperature 866.13 K
Linear burning rate 64.61 mm/s
Mass decomposition rate $107.19 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.42\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2 \mu\text{m}$
Average kinetic flame density 7.04 kg/m^3
Average kinetic flame temperature $2,286.56 \text{ K}$
Binder sublimation heat flux density $1.42\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.021 W/m^2

Secondary Conditional Fuel

Surface temperature 756.12 K
Linear burning rate 6.26 mm/s
Mass decomposition rate $10.38 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.07\text{e}+07 \text{ W/m}^2$
Kinetic flame height $15 \mu\text{m}$
Average kinetic flame density 10.31 kg/m^3
Average kinetic flame temperature $1,560.56 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $121,035.15 \text{ W/m}^2$
Kinetic flame height $689.78 \mu\text{m}$
Average kinetic flame density 13.72 kg/m^3
Average kinetic flame temperature $1,173.56 \text{ K}$
Heat flux density due to metal combustion, $2.69\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $4.91\text{e}-06 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $8.15\text{e}-05 \text{ m}$
Pore diameter $2.83\text{e}-08 \text{ m}$
Diffusion flame heat flux density $2.83\text{e}+06 \text{ W/m}^2$
Diffusion flame height $103 \mu\text{m}$
Heat flux density to surface within skeletal structure $527,616.21 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $8.71\text{e}+06 \text{ W/m}^2$
Total heat flux density to surface $1.21\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.21\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0022 W/m^2

At pressure 6.5 MPa

Experimental burning rate 18 mm/s

Calculated burning rate 19.61 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 118.19 mm/s
Mass decomposition rate $196.08 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$
Kinetic flame heat flux density to surface $2.7 \times 10^8 \text{ W/m}^2$
Kinetic flame height $1.04 \text{ } \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $2.7 \times 10^8 \text{ W/m}^2$
Surface heat balance error -0.015 W/m^2

Secondary Conditional Fuel

Surface temperature 767.67 K
Linear burning rate 8.25 mm/s
Mass decomposition rate $13.68 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.55 \times 10^7 \text{ W/m}^2$
Kinetic flame height $10.3 \text{ } \mu\text{m}$
Average kinetic flame density 16.47 kg/m^3
Average kinetic flame temperature 1,566.34 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $155,145.8 \text{ W/m}^2$
Kinetic flame height $530.68 \text{ } \mu\text{m}$
Average kinetic flame density 21.88 kg/m^3
Average kinetic flame temperature 1,179.34 K
Heat flux density due to metal combustion, $3.47 \times 10^6 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$
Radiative thermal conductivity $2.96 \times 10^{-6} \text{ W/(m} \cdot \text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943 \times 10^{-8}$
Skeleton layer thickness $6.18 \times 10^{-5} \text{ m}$
Pore diameter $1.63 \times 10^{-8} \text{ m}$

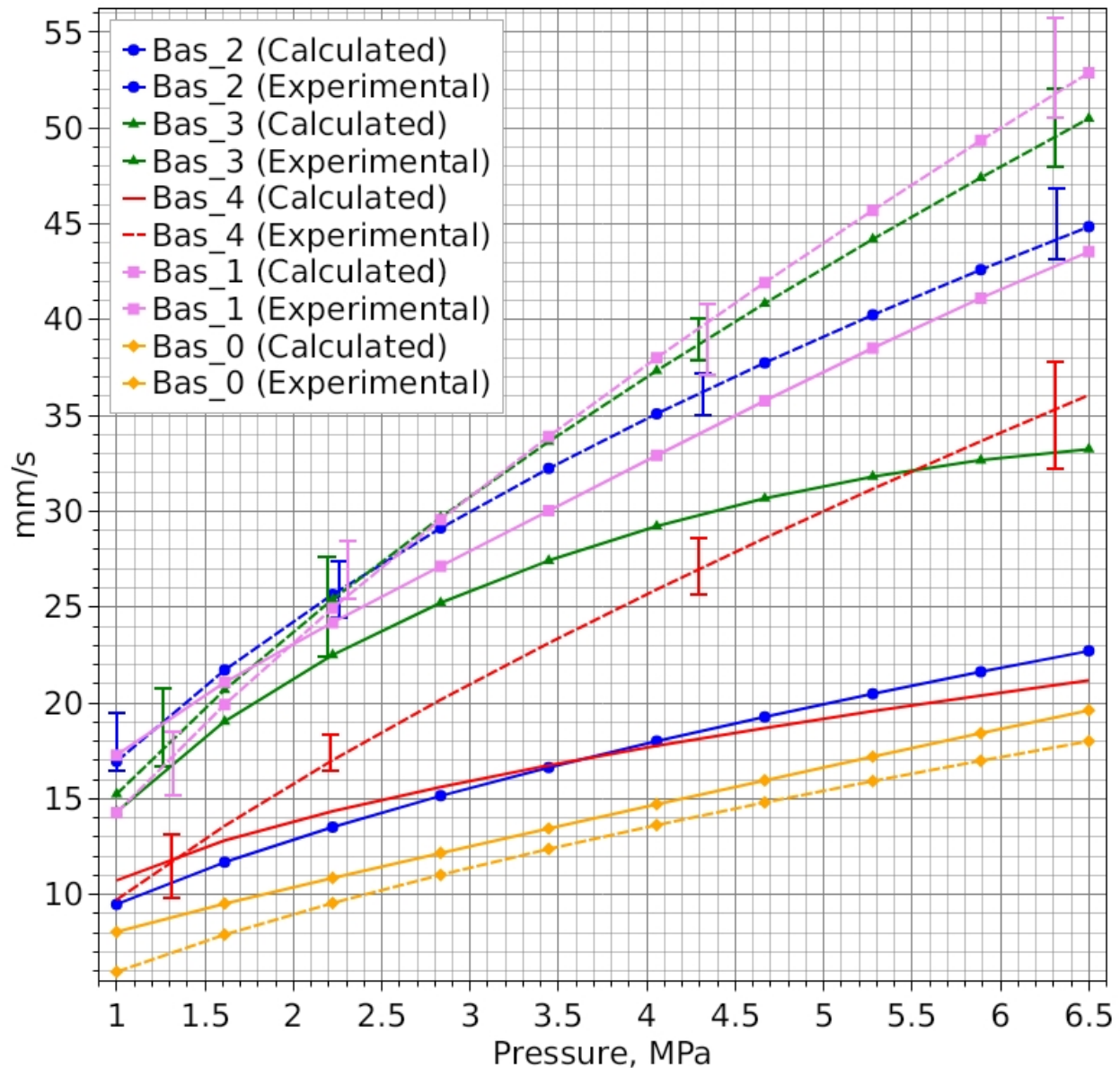
Diffusion flame heat flux density $2.13 \times 10^6 \text{ W/m}^2$
Diffusion flame height $135.83 \text{ } \mu\text{m}$
Heat flux density to surface within skeletal structure $457,646.4 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.36 \times 10^7 \text{ W/m}^2$
Total heat flux density to surface $1.61 \times 10^7 \text{ W/m}^2$
Binder sublimation heat flux density $1.61 \times 10^7 \text{ W/m}^2$
Surface heat balance error 0.0025 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	9.49 mm/s		14.32 mm/s		10.74 mm/s		17.28 mm/s		8.05 mm/s	
Difference	7.45 mm/s		0.93 mm/s		-1.01 mm/s		-3.01 mm/s		-2.08 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	10.72 mm/s	4.59 mm/s	10.72 mm/s	10.72 mm/s	10.72 mm/s	3.8 mm/s	10.08 mm/s	10.08 mm/s	10.82 mm/s	3.78 mm/s
Surface Temperature	757.25 K	722.84 K	757.25 K	757.25 K	757.25 K	715.64 K	600 K	600 K	779.36 K	735.94 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.07e+07 W/m ²		2.07e+07 W/m ²		2.07e+07 W/m ²		1.55e+07 W/m ²		2.15e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	7.12e+06 W/m ²		2.33e+07 W/m ²		2.37e+06 W/m ²		6.72e+06 W/m ²		47,097.32 W/m ²	
Kinetic Flame Height (Skeleton)	12.19 μm		8.01 μm		22.15 μm		14.74 μm		1,815.52 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.15e+06 W/m ²		925,726.33 W/m ²		58,393.66 W/m ²		945,843.51 W/m ²		2.62e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	76.36 μm		173.67 μm		558.9 μm		186.61 μm		62.25 μm	
Diffusion Flame Heat Flux Density	3.82e+06 W/m ²		1.2e+07 W/m ²		2.58e+06 W/m ²		1.85e+06 W/m ²		4.71e+06 W/m ²	
Diffusion Flame Height	75.51 μm		24.22 μm		111.21 μm		165.97 μm		62.22 μm	
Metal Combustion Heat Flux Density	1.99e+06 W/m ²		2.41e+06 W/m ²		5.84e+06 W/m ²		2.79e+07 W/m ²		1.69e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	18 mm/s		29.22 mm/s		17.75 mm/s		32.92 mm/s		14.71 mm/s	
Difference	17.09 mm/s		8.12 mm/s		8.17 mm/s		5.09 mm/s		-1.08 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	63.94 mm/s	7.73 mm/s	63.94 mm/s	14.63 mm/s	63.94 mm/s	5.23 mm/s	59.48 mm/s	14.88 mm/s	64.61 mm/s	6.26 mm/s
Surface Temperature	841.44 K	743.62 K	841.44 K	770.67 K	841.44 K	727.95 K	666.02 K	613.35 K	866.13 K	756.12 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.37e+08 W/m ²		1.37e+08 W/m ²		1.37e+08 W/m ²		1.01e+08 W/m ²		1.42e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.68e+07 W/m ²		6.73e+07 W/m ²		6.84e+06 W/m ²		5.29e+07 W/m ²		121,035.15 W/m ²	
Kinetic Flame Height (Skeleton)	5.04 μm		2.75 μm		7.5 μm		1.85 μm		689.78 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	8.66e+06 W/m ²		4.6e+06 W/m ²		286,682.17 W/m ²		4.35e+06 W/m ²		1.07e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	18.73 μm		34.68 μm		109.55 μm		40.28 μm		15 μm	
Diffusion Flame Heat Flux Density	2.25e+06 W/m ²		8.79e+06 W/m ²		1.86e+06 W/m ²		1.25e+06 W/m ²		2.83e+06 W/m ²	
Diffusion Flame Height	127.31 μm		33.05 μm		153.06 μm		245.08 μm		103 μm	
Metal Combustion Heat Flux Density	3.24e+06 W/m ²		3.2e+06 W/m ²		7.87e+06 W/m ²		4.04e+07 W/m ²		2.69e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	22.71 mm/s		33.25 mm/s		21.17 mm/s		43.54 mm/s		19.61 mm/s	
Difference	22.14 mm/s		17.25 mm/s		14.9 mm/s		9.34 mm/s		-1.61 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	116.92 mm/s	9.6 mm/s	116.92 mm/s	15.63 mm/s	116.92 mm/s	6.1 mm/s	108.36 mm/s	19.13 mm/s	118.19 mm/s	8.25 mm/s
Surface Temperature	874.3 K	752.58 K	874.3 K	773.57 K	874.3 K	734.02 K	691.75 K	622.26 K	900 K	767.67 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.6e+08 W/m ²		2.6e+08 W/m ²		2.6e+08 W/m ²		1.92e+08 W/m ²		2.7e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	2.16e+07 W/m ²		1e+08 W/m ²		9.34e+06 W/m ²		9.66e+07 W/m ²		155,145.8 W/m ²	
Kinetic Flame Height (Skeleton)	3.88 μm		1.85 μm		5.43 μm		1 μm		530.68 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.33e+07 W/m ²		8.2e+06 W/m ²		467,595.95 W/m ²		6.46e+06 W/m ²		1.55e+07 W/m ²	
Kinetic Flame Height (Out-Skeleton)	12.13 μm		19.41 μm		65.86 μm		26.97 μm		10.3 μm	
Diffusion Flame Heat Flux Density	1.8e+06 W/m ²		8.22e+06 W/m ²		1.59e+06 W/m ²		966,451.27 W/m ²		2.13e+06 W/m ²	
Diffusion Flame Height	158.08 μm		35.3 μm		178.47 μm		314.94 μm		135.83 μm	
Metal Combustion Heat Flux Density	3.96e+06 W/m ²		3.4e+06 W/m ²		9.08e+06 W/m ²		5.13e+07 W/m ²		3.47e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341\text{E-}05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

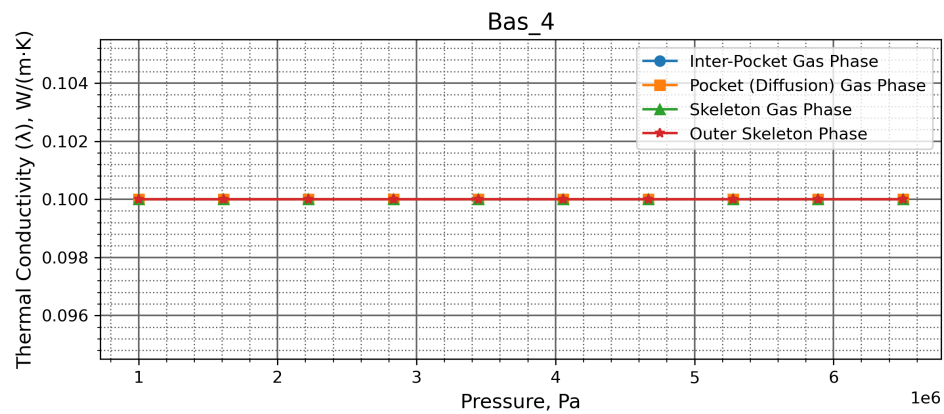
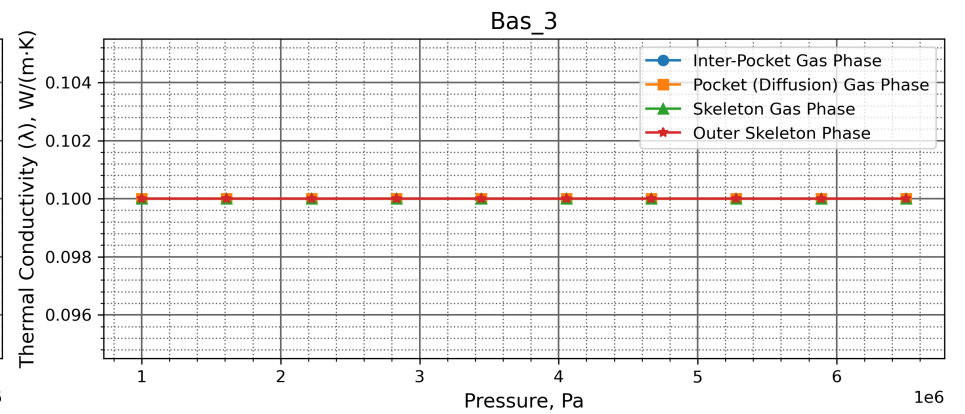
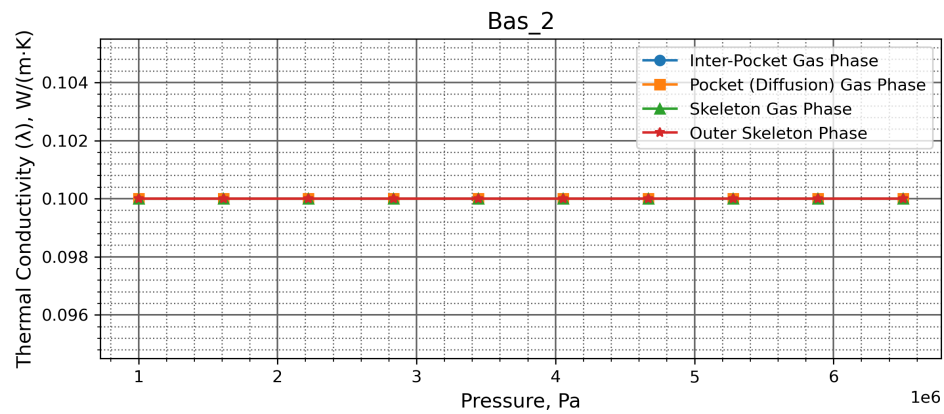
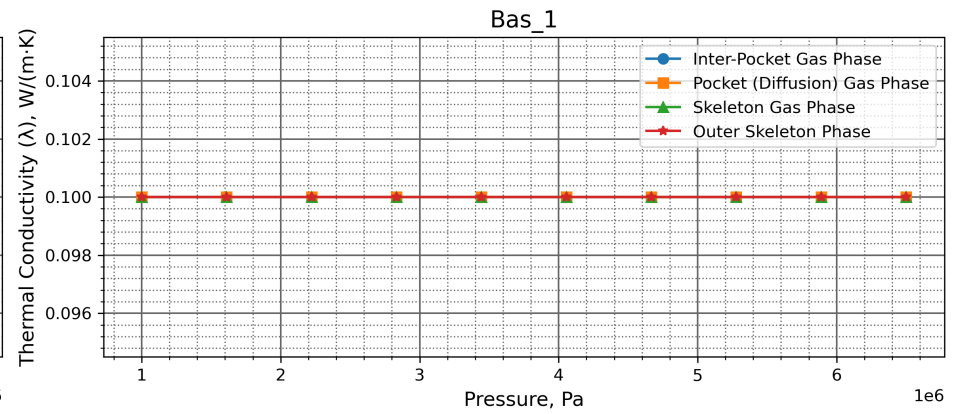
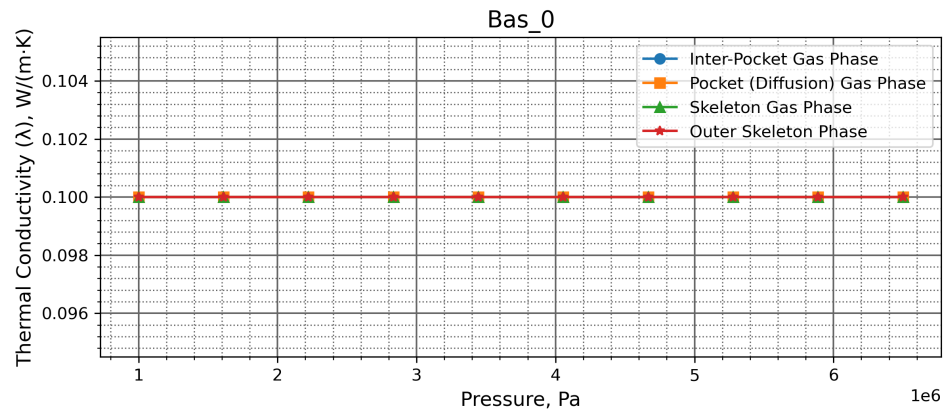
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

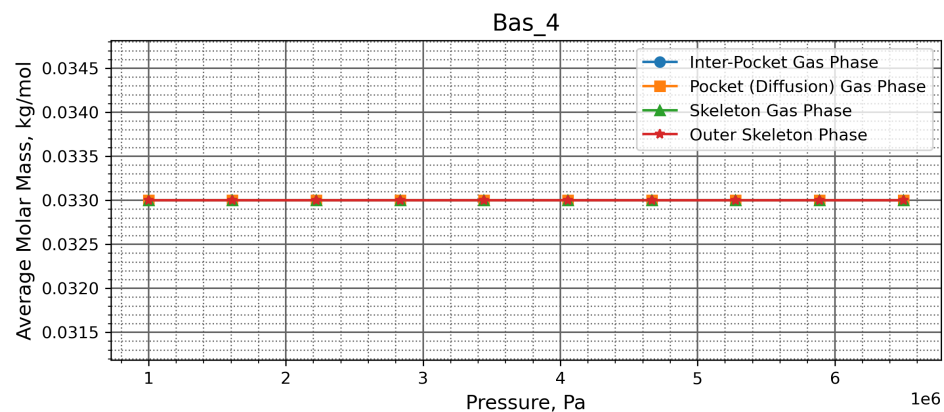
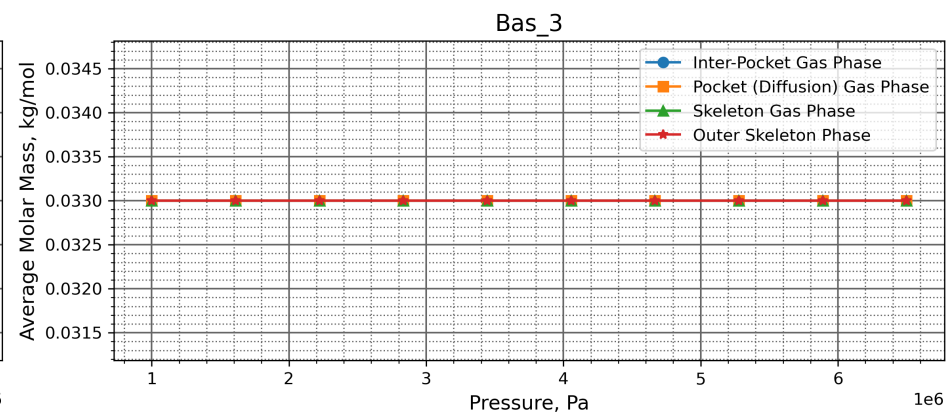
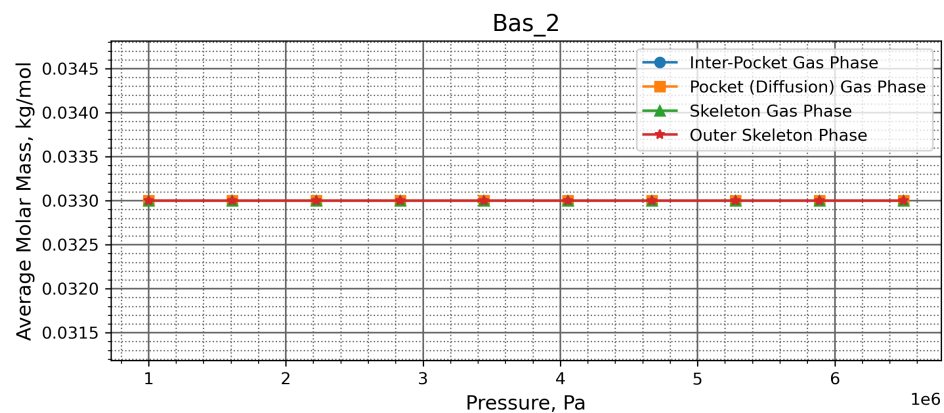
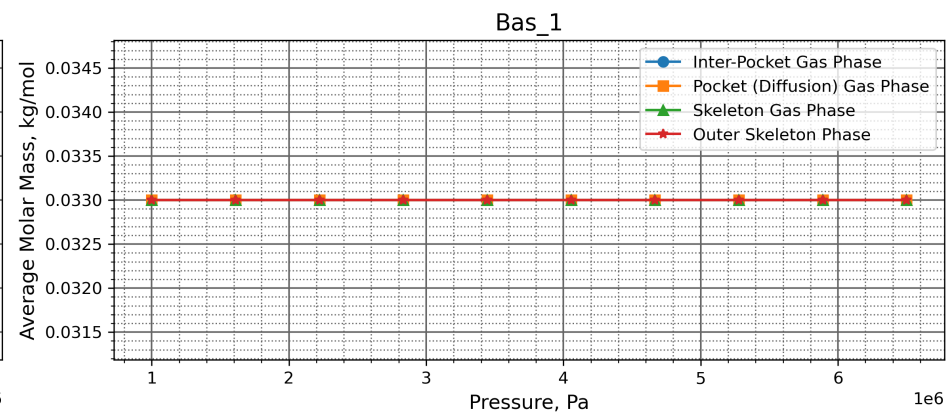
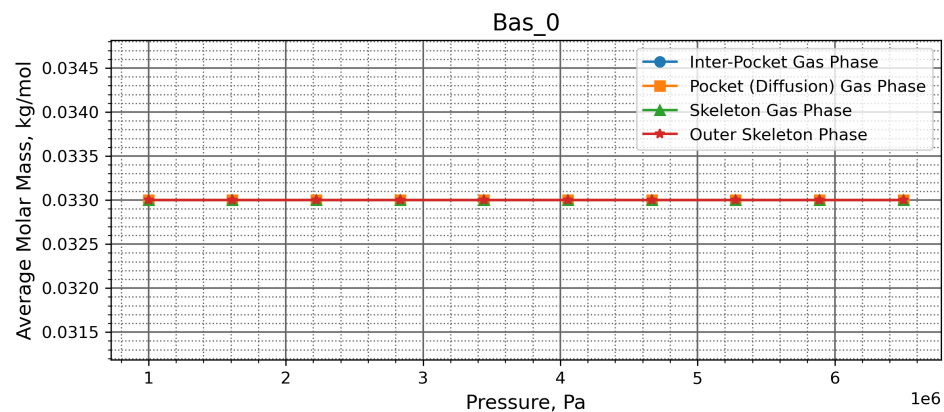
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

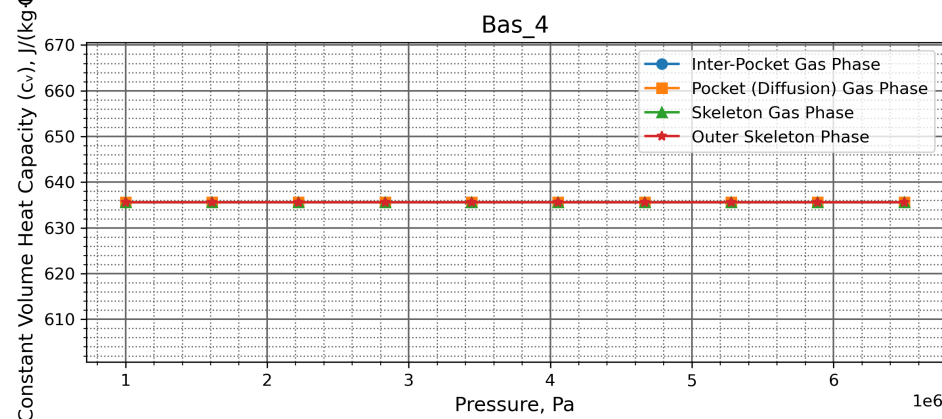
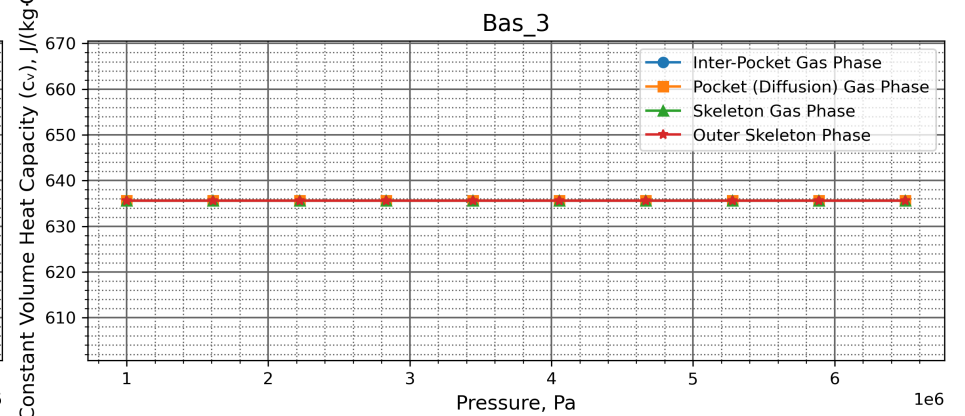
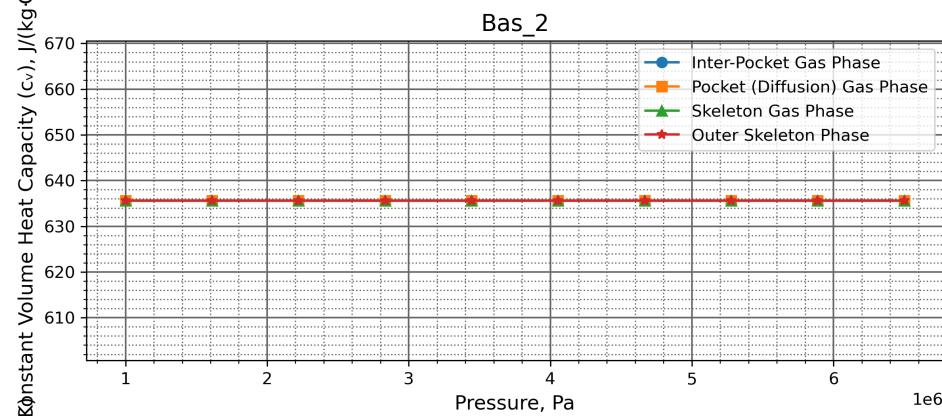
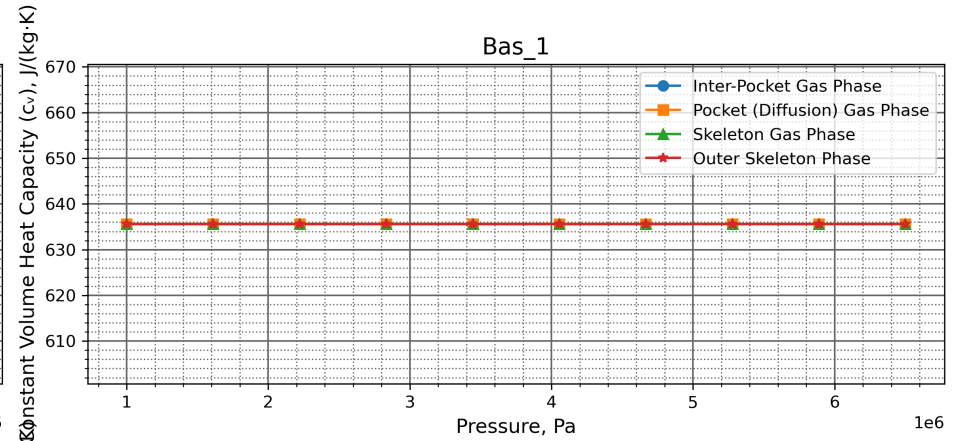
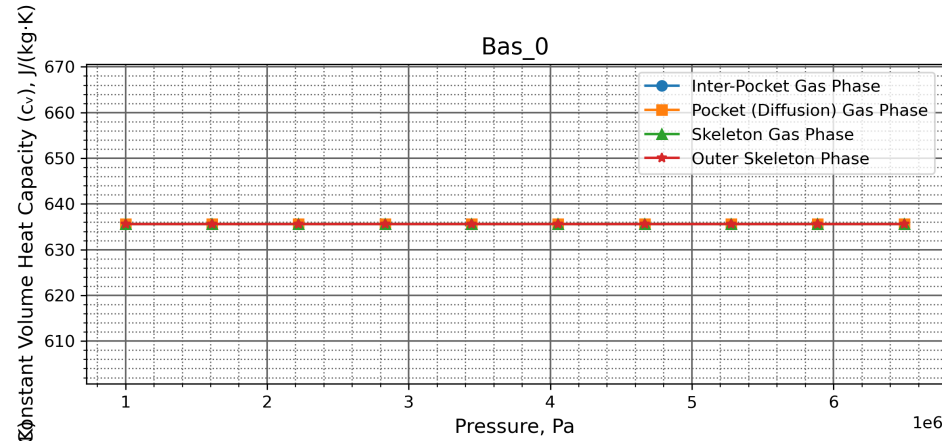
Thermal Conductivity (λ), W/(m·K)



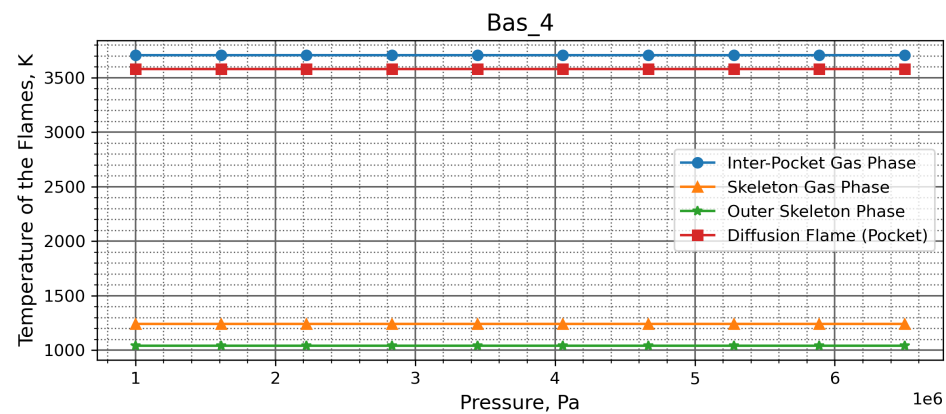
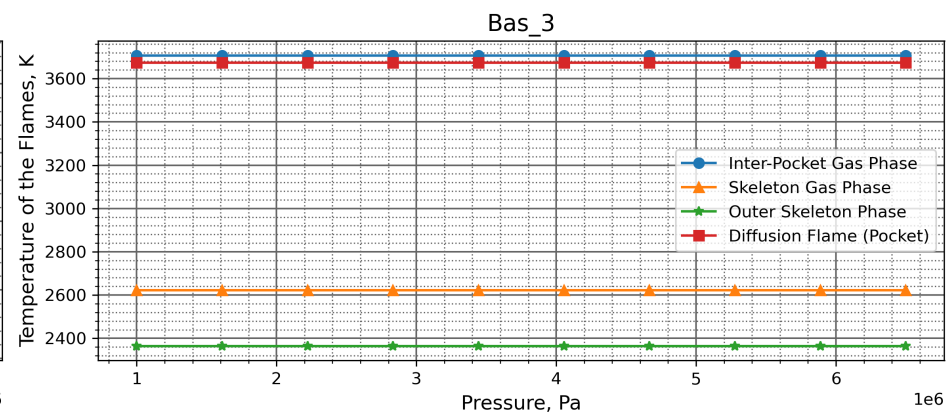
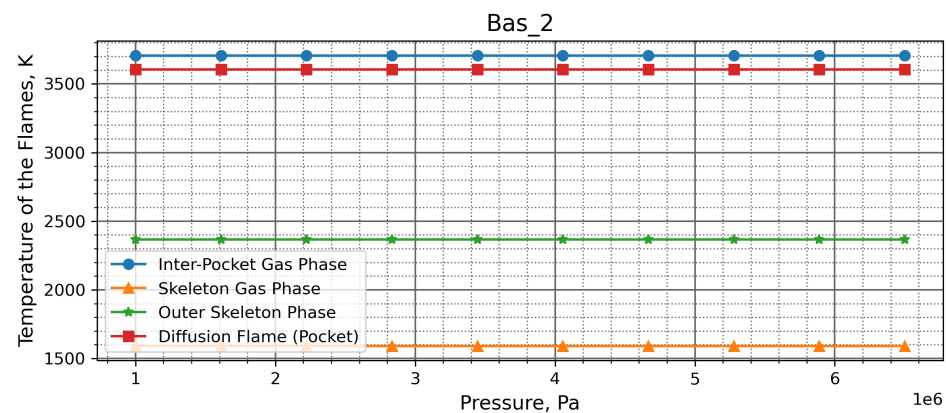
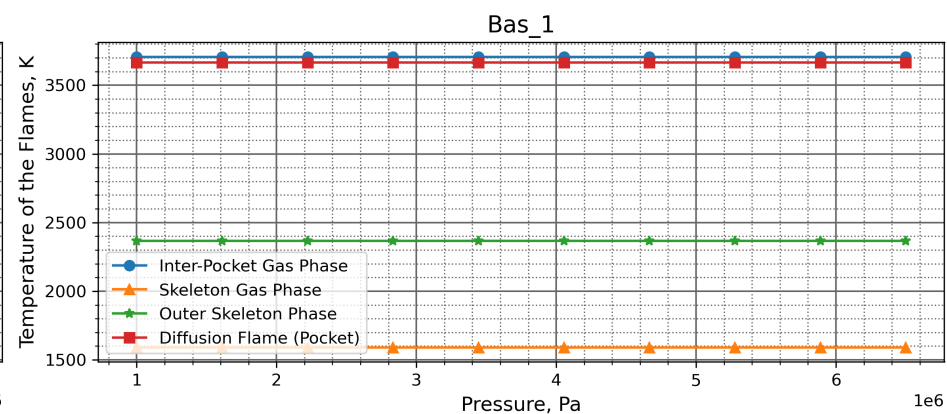
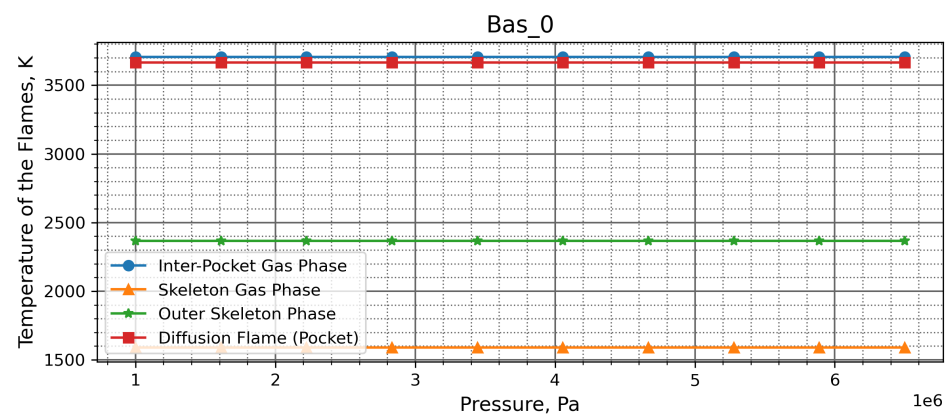
Average Molar Mass, kg/mol



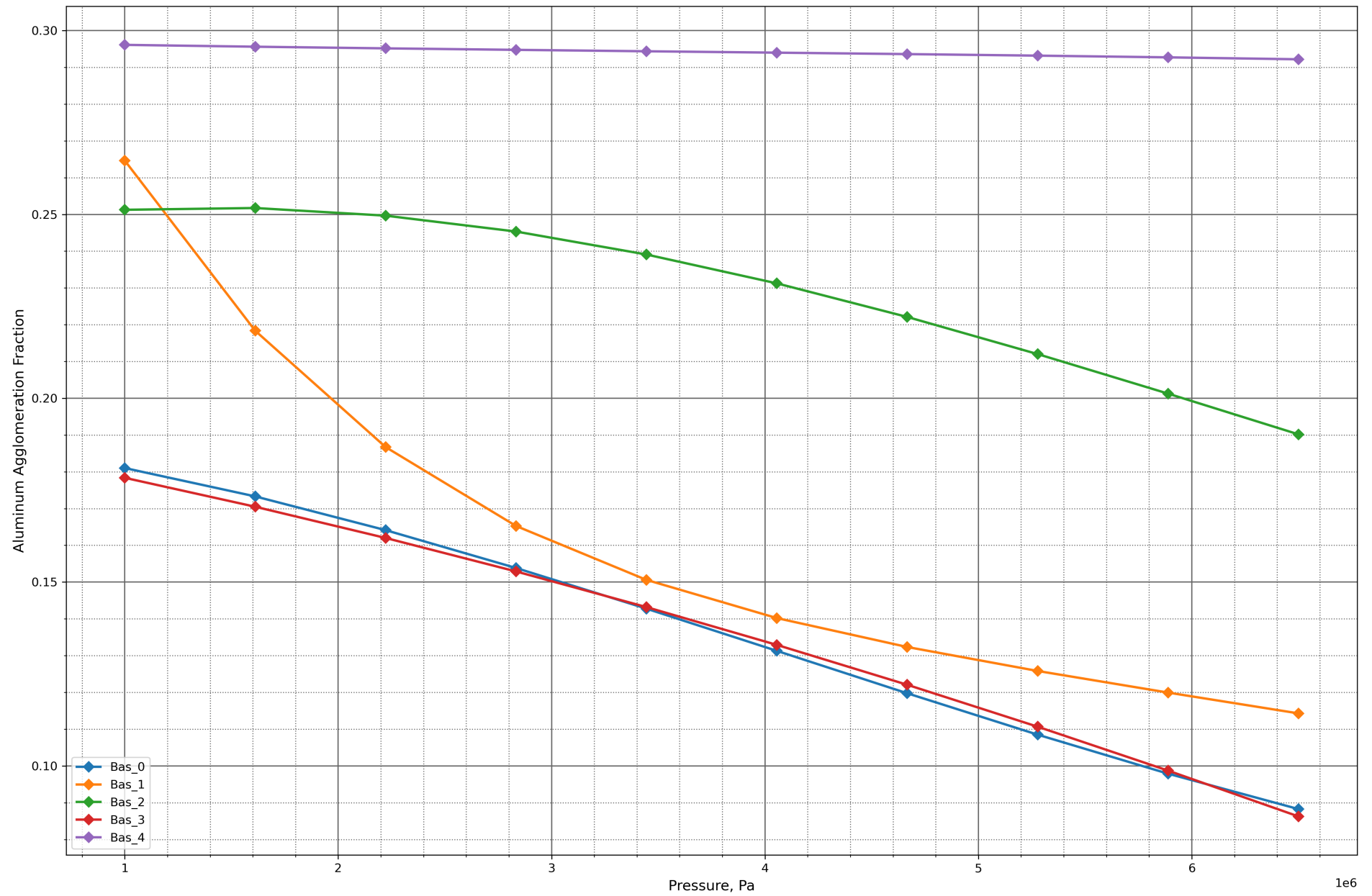
Constant Volume Heat Capacity (c_v), J/(kg·K)



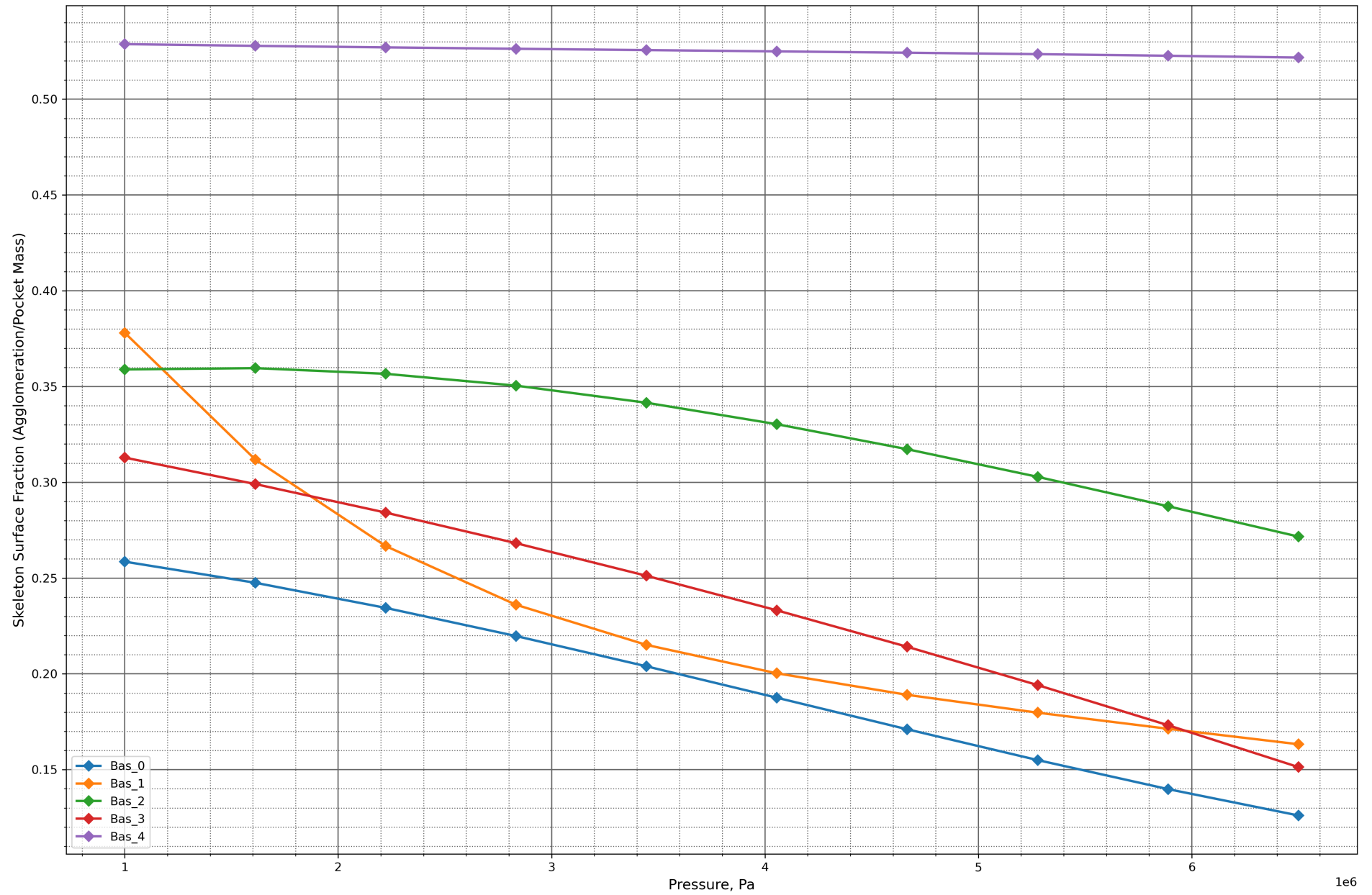
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.117 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.8 (.NET 10.0.8)
Architecture: X64 / X64

Total Execution Time

Total execution time: 01:28:13.587

Garbage Collection Information

Total Memory: 6619032 bytes
GC Pause Time Percentage: 0.14%
Generation 0 Collections: 6235
Generation 1 Collections: 527
Generation 2 Collections: 53

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 3783564

Maximum execution time: 38.097 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8032537038887174 ms

Execution time standard deviation: 0.25960332403153064 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 3423176

Maximum execution time: 5.7405 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.1880294099689921 ms

Execution time standard deviation: 0.14158227156419256 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 2188833

Maximum execution time: 11.9397 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.274429413253548 ms

Execution time standard deviation: 0.06116844238423048 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 23.4701 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.80358906644673 ms

Execution time standard deviation: 0.25581446232905997 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 3423828

Maximum execution time: 17.3506 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.1875029261984039 ms

Execution time standard deviation: 0.14130933540918936 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 2189514

Maximum execution time: 8.3529 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.27361110374264136 ms
Execution time standard deviation: 0.05932454755571652 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 24.4125 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8013694623306844 ms
Execution time standard deviation: 0.25867783753833257 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 3423964
Maximum execution time: 27.9445 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18717757082142908 ms
Execution time standard deviation: 0.1422584612313447 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 2188846
Maximum execution time: 6.9061 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2731835630281906 ms
Execution time standard deviation: 0.061305485368990456 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 8.9959 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8007559503117289 ms
Execution time standard deviation: 0.25544668955033384 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 3422421
Maximum execution time: 7.3245 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.187168968370642 ms
Execution time standard deviation: 0.1407760503196912 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 2188873
Maximum execution time: 17.0398 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2732174566089737 ms
Execution time standard deviation: 0.0608362165494677 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 14.2267 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8042633567792216 ms
Execution time standard deviation: 0.2552644610176571 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 3422873
Maximum execution time: 6.5048 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1882085114463976 ms
Execution time standard deviation: 0.14127444925887808 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 2188194

Maximum execution time: 7.8591 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2752288309446294 ms

Execution time standard deviation: 0.05879326742165819 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 22.8812 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7968634058384655 ms

Execution time standard deviation: 0.2574229855794497 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 3423597

Maximum execution time: 9.5192 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18604785463944493 ms

Execution time standard deviation: 0.14051113054569125 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 2187737

Maximum execution time: 17.0045 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2721906123542192 ms

Execution time standard deviation: 0.06157609336339082 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 18.3206 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8152252662839274 ms

Execution time standard deviation: 0.2557414589469233 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 3423736

Maximum execution time: 12.0343 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.190742526146887 ms

Execution time standard deviation: 0.14291573678151315 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 2188837

Maximum execution time: 11.6679 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2780801560371909 ms

Execution time standard deviation: 0.05851983078459344 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 19.6595 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.802649498384963 ms

Execution time standard deviation: 0.25581232364622114 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 3424273

Maximum execution time: 5.2787 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18716832571468758 ms
Execution time standard deviation: 0.14082477422742767 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 2186608
Maximum execution time: 13.3234 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.273618622359392 ms
Execution time standard deviation: 0.05997801772718336 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 10.3357 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.801673251682362 ms
Execution time standard deviation: 0.25595233978786924 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 3423832
Maximum execution time: 6.7763 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18701409543459546 ms
Execution time standard deviation: 0.14069641121614695 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 2186515
Maximum execution time: 10.6458 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27317794220482183 ms
Execution time standard deviation: 0.05914789979679796 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 11.9312 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7985482225534869 ms
Execution time standard deviation: 0.25638983963185835 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 3423413
Maximum execution time: 13.7669 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1868234963762643 ms
Execution time standard deviation: 0.14067149310695684 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 2189488
Maximum execution time: 10.4128 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2728929609114089 ms
Execution time standard deviation: 0.06018798790473194 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 15.2007 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.798709381446279 ms

Execution time standard deviation: 0.2575085849849179 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 3423430

Maximum execution time: 8.9707 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18643042711549104 ms

Execution time standard deviation: 0.14036577363007352 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 2186916

Maximum execution time: 8.0006 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27254630136682967 ms

Execution time standard deviation: 0.059761117286776626 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 27.1444 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.7979981180117776 ms

Execution time standard deviation: 0.2562558523898951 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 3423475

Maximum execution time: 11.5921 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18663108847004217 ms

Execution time standard deviation: 0.14058225155948276 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 2189535

Maximum execution time: 8.6654 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2724528023530228 ms

Execution time standard deviation: 0.05925225479636408 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 38.3398 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8093107083458871 ms

Execution time standard deviation: 0.2548818939063319 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 3422513

Maximum execution time: 5.3213 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18914543360391825 ms

Execution time standard deviation: 0.1413355977218316 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 2188463

Maximum execution time: 7.4463 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2762081173864979 ms

Execution time standard deviation: 0.05734786364016814 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 17.3943 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8114965664864024 ms
Execution time standard deviation: 0.25591536194240466 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 3424397
Maximum execution time: 5.5234 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1893036722669717 ms
Execution time standard deviation: 0.14176785811495182 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 2187490
Maximum execution time: 12.4399 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27759163127605685 ms
Execution time standard deviation: 0.05914153086958067 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 38.4295 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7958473716291765 ms
Execution time standard deviation: 0.25862255545491386 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 3423546
Maximum execution time: 27.3509 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18615412826350275 ms
Execution time standard deviation: 0.14104153475622852 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 2189992
Maximum execution time: 9.2248 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27119797350857816 ms
Execution time standard deviation: 0.060237633755859124 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 3783180
Maximum execution time: 21.8775 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8076603029726698 ms
Execution time standard deviation: 0.2546038691655441 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 3423331
Maximum execution time: 14.9653 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18945404224716947 ms
Execution time standard deviation: 0.1421544166963507 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 2185845
Maximum execution time: 5.3598 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2751432110693991 ms
Execution time standard deviation: 0.057024847971553876 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 20.3071 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7922471870753778 ms
Execution time standard deviation: 0.25854870524724377 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 3423436
Maximum execution time: 13.2104 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18475395158548374 ms
Execution time standard deviation: 0.14031569746806227 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 2187410
Maximum execution time: 10.3845 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2698089942443355 ms
Execution time standard deviation: 0.06262922143273786 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 55.6646 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8049574631659032 ms
Execution time standard deviation: 0.25732319250266544 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 3423821
Maximum execution time: 4.3349 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1875488832506119 ms
Execution time standard deviation: 0.14060224285226047 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 2185891
Maximum execution time: 9.2574 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2740849716660214 ms
Execution time standard deviation: 0.05813808122542061 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 20.1633 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7986070178262463 ms
Execution time standard deviation: 0.25699186618127406 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 3423229
Maximum execution time: 13.804 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.1866125292231368 ms
Execution time standard deviation: 0.14079747595516398 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 2185438
Maximum execution time: 12.9819 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2728310998069968 ms
Execution time standard deviation: 0.060156735182563016 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 13.4623 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.800614688727534 ms
Execution time standard deviation: 0.2560884323656452 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 3423443
Maximum execution time: 6.3508 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18691950825529435 ms
Execution time standard deviation: 0.14037557740085194 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 2186479
Maximum execution time: 9.8187 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2732060314780205 ms
Execution time standard deviation: 0.058993251804790064 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 33.9362 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8076463740293099 ms
Execution time standard deviation: 0.257085236536774 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 3423043
Maximum execution time: 15.5202 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18838813932517198 ms
Execution time standard deviation: 0.1416864263731815 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 2186284
Maximum execution time: 13.3922 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2755643670264259 ms
Execution time standard deviation: 0.05884966497499526 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 22.5984 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8005544964290316 ms
Execution time standard deviation: 0.2560357080823701 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 3424266
Maximum execution time: 6.2548 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.18724216696366883 ms
Execution time standard deviation: 0.14065577336845989 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 2188060
Maximum execution time: 9.2869 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27320554568885774 ms
Execution time standard deviation: 0.06009704553648147 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 17.0213 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8050312298119776 ms
Execution time standard deviation: 0.2545757993950633 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 3423588
Maximum execution time: 4.4658 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18781211880635282 ms
Execution time standard deviation: 0.1407324281282858 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 2186204
Maximum execution time: 8.803 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27538167000886365 ms
Execution time standard deviation: 0.059447585866383386 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 10.9135 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.80117900670863 ms
Execution time standard deviation: 0.2556132749114864 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 3423353
Maximum execution time: 11.1995 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18708199966525216 ms
Execution time standard deviation: 0.1405606957999991 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 2188159
Maximum execution time: 19.7564 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2734196530508011 ms
Execution time standard deviation: 0.06092380072250086 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 22.835 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8059568993544408 ms
Execution time standard deviation: 0.25467156001191243 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 3423223

Maximum execution time: 7.0405 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18885435325129507 ms

Execution time standard deviation: 0.14169127460185252 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 2186757

Maximum execution time: 8.6501 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.274880262141604 ms

Execution time standard deviation: 0.05726549648746342 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 19.0561 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8093210297950661 ms

Execution time standard deviation: 0.25420012297266303 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 3422797

Maximum execution time: 7.1666 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.1887482738532518 ms

Execution time standard deviation: 0.14112841485144656 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 2186559

Maximum execution time: 8.0396 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2758579933127785 ms

Execution time standard deviation: 0.05731431835913422 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 26.1642 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8013134845553866 ms

Execution time standard deviation: 0.2562151348364246 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 3423970

Maximum execution time: 10.4099 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18680287449948202 ms

Execution time standard deviation: 0.1401781732344761 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance

Call count: 2188008

Maximum execution time: 10.7552 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27295519970677756 ms

Execution time standard deviation: 0.058669264291263436 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 31.9133 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7973130978964885 ms
Execution time standard deviation: 0.25894014604379356 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 3423209
Maximum execution time: 6.3225 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18640645750813906 ms
Execution time standard deviation: 0.14040419571864204 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 2188579
Maximum execution time: 20.7256 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27195137657815494 ms
Execution time standard deviation: 0.062493841933839035 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 17.9962 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.7977955083289421 ms
Execution time standard deviation: 0.2575215486242832 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 3423004
Maximum execution time: 16.3295 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18612688217719678 ms
Execution time standard deviation: 0.14047619269284636 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 2186550
Maximum execution time: 13.2663 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2717823149253618 ms
Execution time standard deviation: 0.06087851052471878 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 3783180
Maximum execution time: 13.7446 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8062770218704902 ms
Execution time standard deviation: 0.2538699555247143 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 3423479
Maximum execution time: 24.6903 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.18846805889563967 ms
Execution time standard deviation: 0.1418759324085216 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 2187901
Maximum execution time: 11.8646 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.27553652646989946 ms

Execution time standard deviation: 0.05819263967534622 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 19.125 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8036170914679859 ms

Execution time standard deviation: 0.2562626185658187 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 3423728

Maximum execution time: 10.9696 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.187446552471479 ms

Execution time standard deviation: 0.14108427878625232 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance

Call count: 2188138

Maximum execution time: 5.1172 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.27411882047661323 ms

Execution time standard deviation: 0.05834543644831305 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 3783180

Maximum execution time: 15.164 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8109892305679259 ms

Execution time standard deviation: 0.25313165505952834 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 3423406

Maximum execution time: 7.1544 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.18938089490406884 ms

Execution time standard deviation: 0.14120349711341088 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance

Call count: 2188296

Maximum execution time: 6.2645 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2771766344224244 ms

Execution time standard deviation: 0.055807336188535395 ms